

**Final answers** (record here — graded on the answer; show your work):

1. \_\_\_\_\_ 2. \_\_\_\_\_ 3. \_\_\_\_\_ 4. \_\_\_\_\_ 5. \_\_\_\_\_ 6. \_\_\_\_\_ 7. \_\_\_\_\_  
8. \_\_\_\_\_ 9. \_\_\_\_\_ 10. \_\_\_\_\_ 11. \_\_\_\_\_

## Domain & Range

**Example** (State the domain (and range where asked)):

$f(x) = \frac{1}{x+3}$ : the denominator can't be 0, so  $x \neq -3$ . Domain: all reals except  $-3$ .

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**Directions:** State the domain. For #9 and #10 also state the range.

1.  $f(x) = \frac{1}{x-4}$

6.  $f(x) = \frac{2}{x^2-16}$

2.  $f(x) = \sqrt{x+6}$

7.  $f(x) = \sqrt{2x-8}$

3.  $f(x) = \frac{x}{x^2-9}$

8.  $f(x) = \frac{x+1}{x^2-x-6}$

4.  $f(x) = \frac{1}{x+7}$

9.  $f(x) = x^2 - 5$

5.  $f(x) = \sqrt{x-1}$

10.  $f(x) = |x| + 2$

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**Challenge** (same sheet):

11.  $f(x) = \frac{\sqrt{x-1}}{x-4}$  — give the domain (both traps apply at once).

## Answer Key — fully worked

**Procedure:** Denominator  $\neq 0$  — exclude those  $x$ ; Even root's inside must be  $\geq 0$  — solve it; Otherwise the domain is all reals; Range: think about the lowest or highest output.

1.  $x \neq 4$
2.  $x \geq -6$
3.  $x \neq \pm 3$  (since  $x^2 - 9 = 0$  at  $\pm 3$ )
4.  $x \neq -7$
5.  $x \geq 1$
6.  $x \neq \pm 4$  (since  $x^2 - 16 = 0$  at  $\pm 4$ )
7.  $x \geq 4$  (need  $2x - 8 \geq 0$ )
8.  $x \neq 3, -2$  (denominator  $(x - 3)(x + 2) = 0$ )
9. Domain all reals; range  $y \geq -5$
10. Domain all reals; range  $y \geq 2$
11.  $x \geq 1$  and  $x \neq 4$  (root needs  $x \geq 1$ ; denominator excludes 4)